

SAP IDENTITY MANAGEMENT (SAP IDM)

Chapter 2 – SAP IDM Repository, Runtime Engine, Passes & Jobs (Architect Level)

Q157. Explain the SAP IDM Repository.

Interviewer's Objective

The interviewer wants to know whether you understand how SAP IDM stores and manages identity data.

Executive Answer

The SAP IDM Repository is the central database that stores all identity-related information required for identity lifecycle management.

It contains:

- Identity records
- Privileges
- Roles
- Tasks
- Workflow information
- Provisioning status
- Audit logs
- Repository configuration

The Runtime Engine continuously reads and updates the repository during provisioning and synchronization.

The repository acts as the single source of truth for SAP IDM operations.

Q158. Explain the Identity Store Schema.

Executive Answer

The Identity Store schema is the logical structure used to organize identity data.

Major object types include:

- MX_PERSON

- **MX_ROLE**
- **MX_PRIVILEGE**
- **MX_GROUP**
- **MX_ASSIGNMENT**
- **MX_TASK**
- **MX_EVENT**

These objects are linked through relationships, allowing SAP IDM to determine who should receive which access.

Typical Flow

HR Record

↓

MX_PERSON

↓

Business Role

↓

Privilege

↓

Provisioning Task

↓

SAP User

Q159. What are Passes in SAP IDM?

One of the most frequently asked technical questions.

Executive Answer

A Pass is a sequence of provisioning tasks executed by the Runtime Engine.

Each pass performs one logical operation.

Examples include:

- **Create User**
- **Modify User**
- **Assign Role**
- **Remove Role**
- **Synchronize Password**
- **Disable Account**

Passes ensure provisioning activities are executed in a controlled and repeatable sequence.

Example

HR Event

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Create Identity Pass

↓

Assign Role Pass

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Provision SAP User Pass

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Provision AD User Pass

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Notification Pass

Q160. What is a Job in SAP IDM?

Executive Answer

A Job is a collection of one or more Passes executed by the Runtime Engine.

Jobs automate identity lifecycle activities.

Examples:

- **Initial Load Job**
- **HR Synchronization Job**
- **Password Synchronization Job**
- **De-provisioning Job**
- **Active Directory Synchronization Job**

A Job may contain multiple Passes executed in a predefined sequence.

Q161. Difference between Job and Pass.

Job	Pass
Collection of tasks	Individual processing step
High-level process	Single operation
Contains multiple passes	Executes one logical activity
Entire workflow	One stage within workflow

Simple Example

Job

Employee Onboarding

↓

Pass 1

Create Identity

↓

Pass 2

Create SAP User



Pass 3

Create AD User



Pass 4

Assign Roles



Pass 5

Send Notification

Q162. Explain Event Handling in SAP IDM.

Executive Answer

SAP IDM is event-driven.

An event initiates processing within the Runtime Engine.

Examples:

- **New employee**
- **Employee transfer**
- **Termination**
- **Password reset**
- **Manager change**
- **Department change**

Each event triggers the appropriate Job and Pass sequence automatically.

Q163. Explain the Runtime Engine Process.

Executive Answer

The Runtime Engine executes provisioning requests using the following process:

Event

↓

Repository

↓

Job

↓

Pass

↓

Task

↓

Provisioning Framework

↓

Target System

↓

Status Update

Every completed activity is recorded in the repository for auditing and monitoring.

Q164. Explain Dispatcher in SAP IDM.

Architect-level interview question.

Executive Answer

The Dispatcher manages communication between the Runtime Engine and connected target systems.

Responsibilities include:

- **Scheduling provisioning requests**
- **Queue management**
- **Load balancing**
- **Error handling**
- **Retry mechanisms**
- **Communication management**

Without the Dispatcher, provisioning tasks cannot be efficiently distributed across systems.

Q165. Explain JavaScript in SAP IDM.

Very common because many implementations use custom scripting.

Executive Answer

JavaScript is used within SAP IDM to implement custom business logic that cannot be achieved through standard configuration.

Typical use cases include:

- **Attribute transformations**
 - **Conditional provisioning**
 - **Data validation**
 - **Custom workflow logic**
 - **Dynamic role assignment**
 - **Repository updates**
 - **String manipulation**
 - **Integration logic**
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Interview Tip

Mention that JavaScript should be modular, well-documented, and used only where standard functionality is insufficient to simplify long-term maintenance.

Q166. Explain SQL Usage in SAP IDM.

This aligns directly with your experience.

Executive Answer

SQL is extensively used for:

- Repository queries
- Reporting
- Data validation
- Custom provisioning logic
- Troubleshooting
- Performance analysis
- Identity reconciliation

Common activities include querying identity records, role assignments, pending tasks, and provisioning history.

Example Repository Tables

Table/View	Purpose
idmv_entry_simple	Identity records
idmv_value_basic_all	Identity attributes
idmv_ovalue_basic_all	Object values
mxi_link	Repository links
mxp_provision	Provisioning information

Q167. How do you troubleshoot failed provisioning?

One of the most valuable production support questions.

Executive Answer

I follow a structured approach:

Step 1

Identify the failed identity.

Step 2

Review Runtime Engine logs.

Step 3

Verify Job execution.

Step 4

Review Pass execution.

Step 5

Validate JavaScript tasks.

Step 6

Check SQL query results.

Step 7

Review Dispatcher queues.

Step 8

Verify connector communication.

Step 9

Validate target system response.

Step 10

Confirm successful provisioning after remediation.

Q168. Performance Optimization in SAP IDM.

Executive Answer

Performance improvements include:

- **Optimizing SQL queries**
- **Reducing unnecessary Passes**
- **Improving Job scheduling**
- **Cleaning obsolete identities**
- **Archiving historical data**
- **Optimizing JavaScript**
- **Monitoring Runtime Engine performance**
- **Tuning repository indexes**
- **Balancing Dispatcher workload**

Performance tuning should focus on both the repository and the processing engine.

Q169. SAP IDM 7.2 vs SAP IDM 8.0.

Very common interview question.

Executive Answer

SAP IDM 7.2

SAP IDM 8.0

Older UI

Eclipse-based Developer Studio

Limited REST integration

Enhanced REST capabilities

SAP IDM 7.2

SAP IDM 8.0

Older provisioning framework Improved provisioning architecture

Traditional workflows Enhanced workflow capabilities

Older Java support Updated Java platform

Basic administration Improved development and monitoring tools

Interview Tip

Highlight that 8.0 introduced improved development tooling, broader integration capabilities, and a more modern architecture while retaining the core identity management concepts.

Q170. Real Production Scenario

Interviewer

"A new employee has been created in HR, but no SAP account is created. How would you troubleshoot?"

Executive Answer

My approach would be:

- 1. Confirm the employee exists in the authoritative HR system.**
- 2. Verify the HR synchronization job completed successfully.**
- 3. Check that the identity was created in the SAP IDM repository.**
- 4. Review Runtime Engine logs for processing errors.**
- 5. Verify the correct Job and Pass sequence executed.**
- 6. Review any JavaScript or SQL tasks involved in identity creation.**
- 7. Validate connector communication with the SAP target system.**
- 8. Check provisioning logs and error messages returned by the connector.**
- 9. Correct the identified issue and rerun provisioning if appropriate.**

10. Confirm that the SAP account is created successfully and document the root cause.

Q171. Real Technical Scenario

Interviewer

"A JavaScript task that assigns business roles is suddenly failing after a transport. How would you investigate?"

Executive Answer

I would approach the issue methodically:

- 1. Review the Runtime Engine logs to identify the failing task and error details.**
- 2. Compare the transported JavaScript with the previous working version to identify changes.**
- 3. Verify that all required repository attributes and constants exist in the target environment.**
- 4. Check whether dependent SQL queries or repository objects return the expected data.**
- 5. Validate that the transport included all related configuration objects, not just the script.**
- 6. Test the logic in a non-production environment using representative data.**
- 7. Correct the issue, perform regression testing, and only then transport the fix to production.**

This minimizes risk while ensuring the integrity of the provisioning process.

Senior Consultant Tips

For SAP IDM interviews:

- Speak in terms of identity lifecycle, not just provisioning.**
- Explain the relationship between Repository → Runtime Engine → Jobs → Passes → Tasks → Target Systems.**
- Emphasize structured troubleshooting rather than trial-and-error.**

- Highlight governance alongside automation.
 - Connect SAP IDM with SAP GRC, Active Directory, and modern identity platforms.
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Chapter Summary

By the end of this chapter, you should be comfortable explaining:

- SAP IDM Repository
 - Identity Store Schema
 - Runtime Engine
 - Jobs
 - Passes
 - Event Handling
 - Dispatcher
 - JavaScript customization
 - SQL usage
 - Performance optimization
 - SAP IDM 7.2 vs. 8.0
 - Production troubleshooting
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Next Chapter – SAP IDM Integration, Active Directory, LDAP, Azure AD, IAS & IPS

This chapter will cover:

- Active Directory integration
- LDAP architecture
- Virtual Directory Server (VDS)
- Microsoft Entra ID integration
- SAP Identity Authentication Services (IAS)
- SAP Identity Provisioning Services (IPS)
- Hybrid identity architectures

- **Cloud identity strategy**
- **SAP IDM migration to SAP Cloud Identity Services and Identity Access Governance (IAG)**
- **Real-world integration and troubleshooting scenarios**

Given your experience with Active Directory, LDAP, VDS, and SAP IDM integrations, this will be another chapter where you can demonstrate expertise beyond what most candidates can offer.